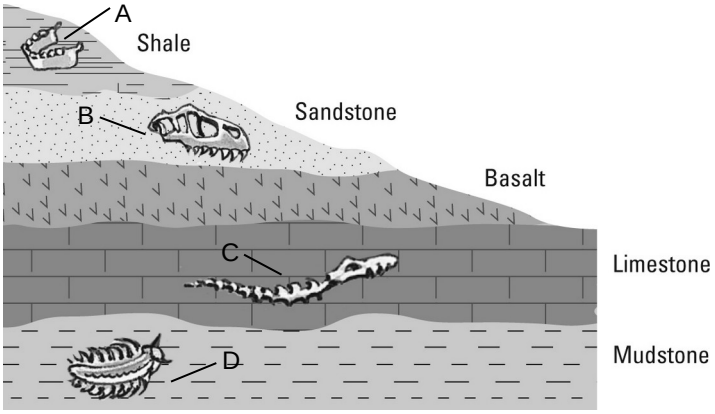
Ages of fossils

Student: ..... Class: .....

Relative ages

1. The following cliff face diagram contains numerous rock layers and fossils.



(a) Identify the youngest and oldest sedimentary rock layers in this sequence.

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(b) Using the code letters (A–D), arrange the fossils in order from oldest to youngest.

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(c) Explain why there are no fossils in the basalt rock layer.

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Radiometric C-14 dating

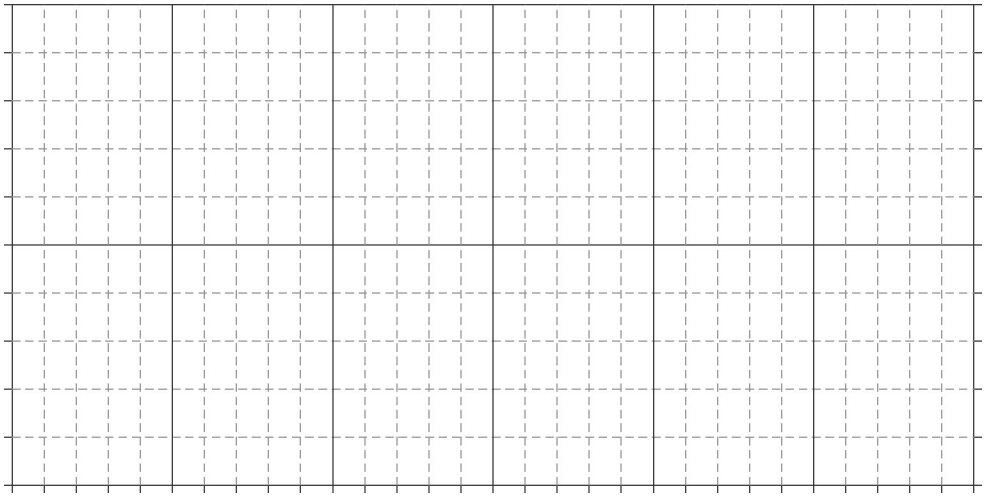
2. The absolute ages of recent carbonaceous fossils can be determined using radiocarbon dating.

The half-life of carbon-14 (a radioactive isotope) is 5730 years. The following table shows how the percentage of C-14 present in a fossil decreases over time.

Time (years)	0	5730	11 460	17 190	22 920	28 650
% C-14 remaining	100	50		12.5	6.25	3.125

(a) Complete the table.

(b) Plot a line graph of this data on the grid below.



(c) The C-14 content of a fossil was found to have decreased by 36% since the organism died. Determine the approximate age of the fossil.

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(d) A fossil is approximately 10 000 years old. What percentage of C-14 still remains?

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(e) Explain why it is difficult to date fossils that are more than 60 000 years old using radiocarbon dating.

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